# OS ORANGE ASSIGNMENT

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Github link: https://github.com/AtharvK-creator/PES2UG24CS092-pes-vcs.git

Screenshots:

Phase 1

1 A) Output of `./test_objects`

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9a
afc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$
```

1 B) Output of `find .pes/objects -type f`

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$
```

Phase 2

2 A) Output of `./test_tree`

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$
```

2 B) Raw binary of tree object

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ ./pes init
Initialized empty PES repository in .pes/
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ echo "hello" > file.txt
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ ./pes add file.txt
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ xxd .pes/objects/2c/f8d83d9ee29543b34a877
27421fdecb7e3f3a183d337639025de576db9ebb4 | head -20
00000000: 626c 6f62 2036 0068 656c 6c6f 0a         blob 6.hello.
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$
```

Phase 3 3 A) pes init -> pes add -> pes status

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ ./pes init
Initialized empty PES repository in .pes/
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ echo "hello" > file1.txt
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ echo "world" > file2.txt
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ ./pes add file1.txt file2.txt
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)

atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$
```

3 B) cat .pes/index output

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ cat .pes/index
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776443631 6 file1.
txt
100664 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776443635 6 file2.
txt
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$
```

Phase 4 4 A) pes log output

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ ./pes log
commit d1c356272de31034efd4c36c5fddfe69aa6f6e1bef590487d935f5a3b0138d09
Author: PES User <pes@localhost>
Date:   1776444054

    Add farewell

commit 47c02e223702d2ab326142ddad1657a0e75e3bd07b80ab6c15ed5e9ef9b9b1a3
Author: PES User <pes@localhost>
Date:   1776444046

    Add worldell

commit 227447071a31ba0b6916fd9cd14aaf717568c1d1217000e4d9c4e6fe048af6c2
Author: PES User <pes@localhost>
Date:   1776443894

    Initial committial commiQ

atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$
```

4 B) `find .pes -type f | sort` showing object growth

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/22/7447071a31ba0b6916fd9cd14aaf717568c1d1217000e4d9c4e6fe048af6c2
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/47/c02e223702d2ab326142ddad1657a0e75e3bd07b80ab6c15ed5e9ef9b9b1a3
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/66/b231ba2ed70d10ccc872a4b00f7ddf539633804c149c1a38af4b90b9fbd6a6
.pes/objects/7d/c3b9d4cee22f645e501f8aaa822f6777b3198434adc74491632478799e9e8b
.pes/objects/d1/c356272de31034efd4c36c5fddfe69aa6f6e1bef590487d935f5a3b0138d09
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/fe/bed41618fe89b49ea9b82684254a1e74aa7f80fdba38fc84b592ccca955267
.pes/refs/heads/main
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$
```

4 C) `cat .pes/refs/heads/main` and `cat .pes/HEAD`

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$ cat .pes/refs/heads/main
cat .pes/HEAD
d1c356272de31034efd4c36c5fddfe69aa6f6e1bef590487d935f5a3b0138d09
ref: refs/heads/main
atharvkulakarni@atharv-pc:~/os-u4-orange-problem/test_phase3$
```

Final Full integration test

```
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 350693d9cd82... Initial commit

Log after first commit:
commit 350693d9cd827717db16d265e9711544f9b11b25fa67ab198c761432efa89084
Author: PES User <pes@localhost>
Date:   1776444318

    Initial commit


--- Second Commit ---
Committed: f0adbb603f7d... Update file.txt

--- Third Commit ---
Committed: b442217e375e... Add farewell

--- Full History ---
commit b442217e375eb17554c941e29dbbaa8894102f075dff2037b2a847223286afc5
Author: PES User <pes@localhost>
Date:   1776444318

    Add farewell
```

```
commit f0adbb603f7d548810b3c115c341ea99dcea3c02208a3c91089119ce008f696a
Author: PES User <pes@localhost>
Date:   1776444318

    Update file.txt

commit 350693d9cd827717db16d265e9711544f9b11b25fa67ab198c761432efa89084
Author: PES User <pes@localhost>
Date:   1776444318

    Initial committial commiQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
b442217e375eb17554c941e29dbbaa8894102f075dff2037b2a847223286afc5

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/35/0693d9cd827717db16d265e9711544f9b11b25fa67ab198c761432efa89084
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/84/c5fed72c2dd9a3130e2e9b98033827f62793fb80d0295ccedf5fc7972d96eb
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/b4/42217e375eb17554c941e29dbbaa8894102f075dff2037b2a847223286afc5
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/de/34fb483a5c16ce29ec3195dd505dfda7e35cc5fe16a99da4b2455b332a93cf
.pes/objects/ec/0af37b159ae92a5850ceaa6e497860729d6158aab2c6833b6068a1e3c5e8ef
.pes/objects/f0/adbb603f7d548810b3c115c341ea99dcea3c02208a3c91089119ce008f696a

=== All integration tests completed ===
atharvkulakarni@atharv-pc:~/os-u4-orange-problem$
```

Phase 5 and 6 Analysis question answers

Q 5.1

Update .pes pointers by verifying the branch file exists at .pes/refs/heads/, reading the target commit hash, and rewriting .pes/HEAD to ref: refs/heads/ using an atomic temp + rename. Resolve the target snapshot by reading the commit object, its root tree, and recursively expanding tree objects to build the full path → blob mapping.

Update the working directory by writing each blob to its file with correct mode, removing tracked files not present in the target tree, and recreating directories as needed. Rebuild the index by setting .pes/index entries to match the target tree with updated stat metadata.

This is complex because it must safely handle deletions, nested directories, permissions, partial failures, and avoid data loss when uncommitted changes overlap with target changes.

Q 5.2

Use a 3-way comparison per tracked file with base as the current HEAD tree blob hash, staged intent as the hash in .pes/index, and working file as stat plus recomputed blob hash. Build the target tree mapping and for each path differing between current HEAD and target, check if the working directory is dirty relative to the current committed state.

If the file is dirty and the target also changes that path, refuse checkout. Also detect untracked file overwrite cases where an untracked working file would be overwritten by the target tree and refuse.

This enforces the rule that checkout should not overwrite local non-committed content.

Q 5.3

In detached HEAD, .pes/HEAD stores a commit hash directly instead of a branch ref, and new commits still work by pointing to the previous detached commit while HEAD moves forward. The risk is that no branch references these commits, making them potentially unreachable.

Recovery involves creating a branch ref pointing to that commit chain or switching to a branch after anchoring the detached commit. If not recovered and no refs point to them, future GC may delete them.

Q 6.1

Use a mark-and-sweep approach where the mark phase initializes a hash set, seeds it with all branch tips, and performs DFS or BFS traversal marking commits, trees, and blobs while enqueueing referenced objects. The sweep phase scans all files in .pes/objects and deletes those whose hashes are not in the reachable set, optionally cleaning empty directories.

A hash set ensures O(1) expected membership checks and deduplication during traversal. In scale, traversal visits unique reachable objects, typically hundreds of thousands to a few million, making the complexity proportional to reachable objects plus total objects on disk.

Q 6.2

A race occurs when commit writes blob and tree objects, then commit object, and finally updates the branch ref while GC runs before the ref update and cannot see the new chain. GC treats new objects as unreachable and deletes them, after which the branch points to a broken history with missing objects.

Real systems avoid this through synchronization and locking around ref updates and GC, reachability protections like reflogs and grace periods, avoiding immediate pruning of recent objects, and ensuring objects are durable before references move.